

Subhadeep Ghorai

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SUMMARY

Software Engineer building AI-agent evaluation systems and full-stack platforms on the MERN stack. Shipped a VS Code extension to the Marketplace using Node.js runtime internals, and a merged documentation fix to **electron/electron**. Experienced in concurrency-safe systems and async execution. Open to SDE / New Grad roles.

EXPERIENCE

Newton School of Technology

Dec 2025 – Present

Software Engineer and SME

- Built AI-agent evaluation pipelines using Node.js and Claude API to autograde GitHub repos for 441 students in ~110 minutes.
- Designed internal full-stack tools (React, Node.js, Express, MongoDB) for evaluation pipelines and operational workflows across the engineering program.
- Delivered 100+ technical lectures on Electron, React Native, System Design, OOP, and Software Engineering to 500+ engineers; 4.46/5 rating (91.55% CSAT).

Codingal

Dec 2023 – Jan 2026

Software Engineer & Senior Coding Instructor

Remote

- Designed and shipped **Atlas**, an operations platform (React, Node.js, Google Sheets API) replacing manual spreadsheet/Canva workflows for 700+ teachers; onboarded 100+ in first 10 days and automated 168+ certificates.
- Mentored 100+ students across 10+ countries through 2,500+ live sessions; promoted to Senior Coding Instructor (4.44/5 rating, 80% renewal rate).

PROJECTS

EventLoop Studio | TypeScript, Node.js vm, Acorn, React, VS Code Extension [Marketplace]

- Published VS Code extension executing JavaScript in a real Node **vm** sandbox with native Promise intrinsics for spec-correct async behavior.
- Built scrubbable step-by-step execution replay engine capturing push/pop, console calls, timer/microtask/phase transitions as reconstructable execution steps.
- Modeled browser and Node.js (libuv phases) execution semantics for accurate cross-runtime visualization.

NSTEP – Internship Evaluation & Slot-Booking Platform | React, Node.js, Express, MongoDB, JWT [Live]

- Engineered concurrency-safe booking system for 300+ students using MongoDB transactions and atomic updates; validated under load (20 simultaneous requests → 5 succeed).
- Implemented cross-admin isolation with JWT-based authorization and ScheduleLock mechanism, eliminating scheduling conflicts by design.
- Kept Google Sheets off critical path with async sync and account-based throttling; covered by 206 unit, concurrency, and integration tests.

Application Development Course Management System | React, Node.js, Express, MongoDB, Authentication

- Built full-stack analytics dashboard tracking 832 projects across 200+ students, 5 admins spanning 2 campuses; live metrics on submissions, engagement, and project health.
- Designed concurrency-safe student enrollment and project assignment system; role-based dashboards (student, instructor, admin) with isolated data views.
- Implemented real-time project status tracking with AI chat history, repository submission monitoring, and problem statement bank management serving two institutions.

OPEN SOURCE

electron/electron – Merged documentation fix (PR #53114) documenting that pending navigation delays `debugger.sendCommand()` resolution (~445ms) with timestamped logs.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java

Frontend Architecture: React, React Native, Electron; state management, component design patterns

Backend & Databases: Node.js, Express.js, MongoDB, PostgreSQL

Cloud & Infrastructure: AWS, Docker, Kubernetes, GitHub Actions, CI/CD pipelines

EDUCATION

VIT-AP University

B.Tech in Electronics and Communication Engineering

September, 2026

CGPA: 8.52/10